

Final Examination

PHY102-Electromagnetism

Date: 25th April 2024, 2pm to 5pm

Q1 Explain the meaning of the equation $\nabla \cdot E = \frac{\rho}{\epsilon_0}$ and equation $\nabla \cdot B = 0$. Mention the fundamental differences between them. [4+2]

Q2 Derive the Hall co-efficient (R_h) expression for a block of semiconductor material (dimensions of sample $[L \times D \times t]$ is L, d, t) where a current of " I ", applied magnetic field " B " and measurement of hall voltage (V_h) are available to you. How can you conclude the current carrier type (hole or electron) in the given semiconductor material? [5+2]

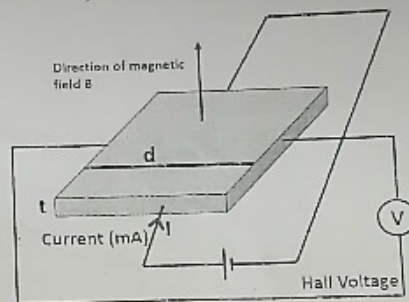


Figure 1: Figure shows a hall experiments.

Q3 what is atomic magnetic dipole moments and magnetization? Draw an ideal magnetic and electric dipoles system. Write-down the names of different types of materials based on the magnetization types. [2+3+2]

Q4 Does the magnetization of a ferro-magnet (such as iron) depend on the applied magnetic field at that instant only? explain. [1+4]

Q5 In magnetized materials, the speed of electron moving around nucleus remain unchanged under applied magnetic field. Prove that the above statement is incorrect. [5]

Q6 Derive the relationship between magnetic susceptibility (χ_m) and permeability (μ) for a ferro-magnetic materials, where applied magnetic field is B and magnetic field intensity inside the material is " H ". [5]

Q7 plot H vs (M) magnetization curve for ferro-magnetic, paramagnetic and diamagnetic materials. what is remanence and coercivity and show its position in the H vs M plot? Which materials will not show these (remanence and coercivity) properties? [6+2+1]

Q8 Draw a circuit diagram by which you can generate E-M wave. Explain the needs of each part of the circuit. Show the oscillation direction of the Magnetic Field (B), Electric Field (E) and the propagation direction of E-M wave. [3+3]